Problem 1:
1. Find the names of employees who live in the same cities and on the same streets as their managers.

```
SELECT e.employee_name
FROM employee e
JOIN manages m ON e.employee_name = m.employee_name
JOIN employee mng ON m.manager_name = mng.employee_name
WHERE e.street = mng.street
AND e.city = mng.city;
```

2. Find the names of employees who earn more than the average salary of all employees of their company.
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```
SELECT w1.employee_name
FROM works w1
JOIN (
   SELECT company_name, AVG(salary) AS avg_salary
   FROM works
   GROUP BY company_name
) AS avg_salaries ON w1.company_name = avg_salaries.company_name
WHERE w1.salary > avg_salaries.avg_salary;
```

Problem 2:
1. Retrieve the last names of all employees in department 5 (i.e. Dnum = 5) who work more than 10 hours on project with Pname = 'ProjectX'.

```
SELECT e.Lname
FROM Employee e
JOIN Works_on w ON e.Ssn = w.Essn
JOIN Project p ON w.Pno = p.Pnumber
WHERE e.Dnum = 5
  AND p.Pname = 'ProjectX'
  AND w.Hours > 10;
```

2. List the last names for all employees who have a dependent with the same first name as Themselves.

```
SELECT e.Lname
FROM Employee e
JOIN Dependent d ON e.Ssn = d.Essn
WHERE e.Fname = d.Dependent_name;
```

3. For each project, list the project name and the total hours (by all employees) spent on that Project.

```
SELECT p.Pname, SUM(w.Hours) AS Total_Hours
FROM Project p
```

```sql
JOIN Works_on w ON p.Pnumber = w.Pno
GROUP BY p.Pname;
```

4. Retrieve the birthday of all employees who work on every project.

```sql
SELECT e.Bdate
FROM Employee e
WHERE NOT EXISTS (
   SELECT p.Pnumber
   FROM Project p
   WHERE NOT EXISTS (
      SELECT w.Pno
      FROM Works_on w
      WHERE w.Essn = e.Ssn
        AND w.Pno = p.Pnumber
   )
);
```

5. List the last names of all department managers who have no dependents.

```sql
SELECT e.Lname
FROM Employee e
JOIN Department d ON e.Ssn = d.Mgr_ssn
LEFT JOIN Dependent dp ON e.Ssn = dp.Essn
WHERE dp.Essn IS NULL;
```

6. Find the last names and addresses of all employees who work on at least one project located in Houston but whose department has no location in Houston.

```sql
SELECT DISTINCT e.Lname, e.Address
FROM Employee e
JOIN Works_on w ON e.Ssn = w.Essn
JOIN Project p ON w.Pno = p.Pnumber
WHERE p.Plocation = 'Houston'
  AND e.Dnum NOT IN (
   SELECT dl.Dnumber
   FROM Dept_locations dl
   WHERE dl.Dlocation = 'Houston'
);
```

Problem 3:

1. Retrieve the names and majors of all ‘A’ students (i.e., students who have a grade of ‘A’ in all their courses).

```sql
SELECT s.Name, s.Major
FROM STUDENT s
WHERE NOT EXISTS (
   SELECT gr.SectionIdentifier
```

```
   FROM GRADE_REPORT gr
   WHERE gr.StudentNumber = s.StudentNumber
   AND gr.Grade <> 'A'
);
```

2. Retrieve the names and majors of all students who do not have a grade of ‘A’ in any of their courses.

```
SELECT s.Name, s.Major
FROM STUDENT s
WHERE NOT EXISTS (
   SELECT gr.SectionIdentifier
   FROM GRADE_REPORT gr
   WHERE gr.StudentNumber = s.StudentNumber
   AND gr.Grade = 'A'
);
```